Kinetic energy and net work

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect net work to a change in kinetic energy.

SUCCESS CRITERIA - I can...

1. Calculate kinetic energy with speed squared.
2. Use net work rather than one force's work.
3. Connect the sign of work to a speed change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "kinetic energy" and "work" to explain your answer. Sentence starters are optional.

kinetic energy: energy associated with motion

work: energy transferred by a force along a displacement

initial: at the beginning of the interval; subscript i

01 Warm-up: bring back an earlier idea

1. Keep mass fixed and multiply speed by **3**.
2. By what factor does kinetic energy change?
3. Explain from the equation.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $K = \frac{1}{2}mv^2$; $W_{net} = K_f - K_i$.
i = initial; f = final. Positive net work increases kinetic energy.

Worked reasoning: A **2 kg** cart at **2 m/s** has K=**4 J**. Adding **5 J** of net work gives final K=**9 J**, so final speed is **3 m/s**.

Your turn: Draw before-and-after energy bars and **label the transfer** between them.

Sentence starter: In my work, I used … to show … . I labeled it … .

Initial **Transfers** **Final**

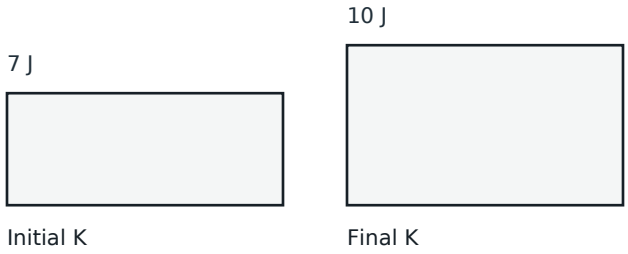
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A lab cart's kinetic energy changes from **7 J** to **10 J**, as shown.
1. **Find the net work**.
2. **Explain how its speed changes** if mass stays constant.

Sentence starter: I think … . I can point to …, which shows …, so … .



Marzano check - circle: ① ② ③

04 Apply the physics

A **2 kg** cart moves at **3 m/s**. A pull and all other forces together do **+7 J** of net work. **Find its initial** and final kinetic energy. **Complete the energy account**.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student adds the work of a pull but ignores friction. Why might their predicted speed be too high?

Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two **4 kg** carts latch together after colliding. Is this perfectly inelastic? **State what remains conserved** if external impulse is negligible.
Use before and after snapshots to organize the momentum account.

Word help: inelastic = a collision that transforms some kinetic energy to other forms

Sentence starter: I notice … . This supports … because … .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

K changes from **20 J** to **8 J**.
1. **Find net work** and state whether speed rises or falls.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. Keep the cart's mass fixed.
2. **Compare the energy needed** to increase speed from **1** to **2 m/s** with that from **2** to **3 m/s**.
3. Use m=**2 kg**.
4. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3